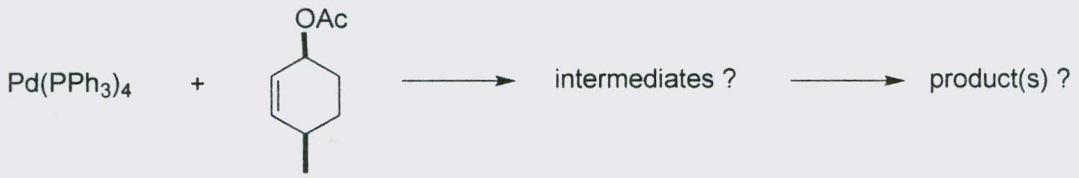
Organic Cumulative Exam March 29, 2013

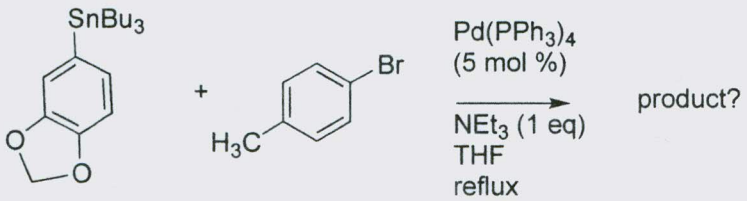
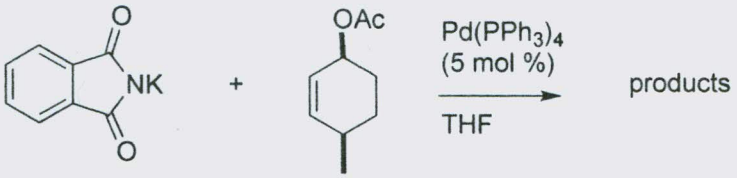
Organometallic Reactions

1. Draw <u>all</u> of the intermediates and product(s) of the <u>stoichiometric</u> reaction shown below. You <u>do not</u> need to show electron pushing arrows.

2. Draw <u>all</u> of the intermediates and product(s), including stereochemistry where appropriate, of the <u>stoichiometric</u> reaction shown below. You <u>do not</u> need to show electron pushing arrows.



3. Draw the major organic product of the following reaction.



4. Two isomeric organic products are possible in the following reaction. Draw both of them, including stereochemistry. Which do you think is the major product, and why?